

Grade 7 Mathematics — Mock Exam — Paper B

MEDIUM • NON-CALCULATOR • 75 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to work without one.
- 75 minutes for 80 marks — about a minute a mark. If a 3-mark question has taken you five minutes, leave it and come back to it.
- No formulas and no methods are given anywhere on this paper — not one. Remembering the right one is part of what is being tested.
- Give every fraction and every ratio in its simplest form, and convert to a common unit before you simplify.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – QUICK FIRE

1. Write down the answer to each of these.

(a) Write 90 as a product of its prime factors, in index form. [1]

(b) -8×-5 [1]

(c) $\sqrt{196}$ [1]

(d) Write $5^6 \div 5^4$ as a single power of 5. [1]

(e) $7.3 \div 0.1$ [1]

2. Write down the answer to each of these.

(a) Round 34 682 to the nearest thousand. [1]

(b) 0.6×0.05 [1]

(c) Write $\frac{5}{9}$ as a decimal. Use dot notation. [1]

(d) Write the ratio 35 : 49 in its simplest form. [1]

(e) Factorise $7x - 21$. [1]

SECTION 2 – SHORT ANSWERS

3. Two numbers are 20 and 30.

(a) Find the highest common factor (HCF) of 20 and 30. [1]

(b) Find the lowest common multiple (LCM) of 20 and 30. [1]

4. Simplify $(3^2)^3 \div 3^4$. Give your answer as an ordinary number. [2]

Deal with the outside power first. A power raised to a power multiplies the indices.

5. Work out:

(a) $0.48 \div 0.01$

[1]

(b) 350×0.01

[1]

6. Write these decimals in ascending order — smallest first.
0.505 0.55 0.5 0.5005

[2]

Give them all the same number of decimal places first. 0.5 is 0.5000.

7. Write these fractions in ascending order — smallest first. You must show your working.
 $\frac{3}{5}$ $\frac{7}{10}$ $\frac{11}{20}$ $\frac{13}{20}$

[2]

8. Work out $5\frac{1}{3} - 2\frac{3}{4}$. Give your answer as a mixed number in its simplest form.

[2]

A third minus three quarters is negative. Either borrow, or use improper fractions.

9. A notebook costs ₹n. A ruler costs ₹5 less than a notebook. Write an expression, in its simplest form, for the total cost of 3 notebooks and 4 rulers.

[2]

10. Expand and simplify $4(2x + 3) + 3(x - 5)$. [2]

SECTION 3 – MULTI-STEP

11. Work out:

(a) -5×-9 [1]

(b) $-5 - 9$ [1]

(c) $72 \div -9$ [1]

(a) and (b) start with the same two numbers and end a long way apart.

12. Work out $5.76 \div 0.24$, showing your method. [3]

13. Work out $8 \times 3\frac{2}{5}$. Give your answer as a mixed number. [3]

14. Work out $12 \div \frac{3}{4}$. [3]

15. $T = 3a + 2b$

(a) Find T when $a = 7$ and $b = -4$. [2]

(b) Find a when $T = 26$ and $b = 4$. [1]

16. Expand and simplify $6(x - 2) - 3(2x - 5)$. [3]

If your answer still has an x in it, check the signs again.

17. Solve $3(x + 4) = 5x - 2$. Show every step. [3]

18. A bicycle costs ₹640. Its price is reduced by 35%. Work out the new price. [3]

SECTION 4 – PROBLEMS

19. ₹4200 is shared between Kiran and Leela in the ratio 2 : 5.

(a) Work out how much each of them receives. [2]

(b) Kiran's share is then decreased by 25%. Work out his new amount. [2]

(c) How much more than Kiran's new amount is Leela's share? [1]

20. Meera has x stickers. Her brother has 8 fewer than three times as many as Meera. Together they have 64 stickers.

(a) Form an equation in x . [2]

(b) Solve your equation to find how many stickers Meera has. [2]

(c) Write down how many stickers her brother has. [1]

21. Two shops sell the same petrol.

Shop A: 2.5 litres for ₹245. Shop B: 4 litres for ₹380.

(a) Work out the price per litre at each shop. [3]

(b) Write down which shop is better value. [1]

(c) Work out the cost of 10 litres at the better-value shop. [1]

22. A rectangular floor measures 60 cm by 84 cm. It is to be covered completely by identical square tiles, with no tile cut and no gaps.

(a) Work out the largest possible side length of a tile. [3]

(b) Work out how many of these tiles are needed. [2]

The tile has to fit a whole number of times along both sides.

23. A phone costs ₹1500. In one week its price is increased by 20%. The following week the new price is reduced by 20%.

(a) Work out the price after the increase. [1]

(b) Work out the price after the reduction. [2]

(c) Work out the overall percentage change from ₹1500. [2]

The answer to (b) is not 1500. If you think a 20% rise and a 20% fall cancel, this question is here to argue with you.

24. A rectangle has length $(2x - 1)$ cm and width 5 cm.

(a) Write an expression for its area. Expand your answer. [2]

(b) Write an expression for its perimeter, in its simplest form. [2]

(c) Factorise your expression for the perimeter. [1]

End of paper. Before you hand it in, check every answer that came out as a round, comfortable number — those are the ones a wrong method produces most often.